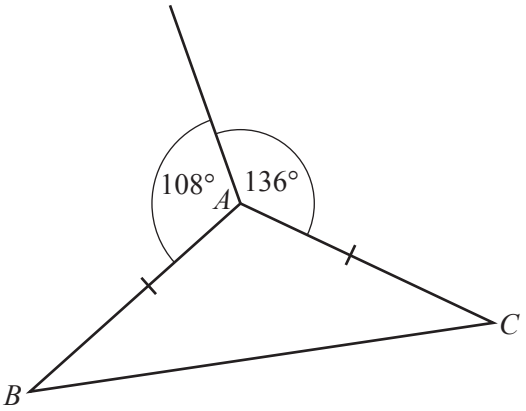
0580_s18_qp_32_Q8

8 **(a)**



NOT TO SCALE

In the diagram, $AB = AC$.

Find

(i) angle BAC,

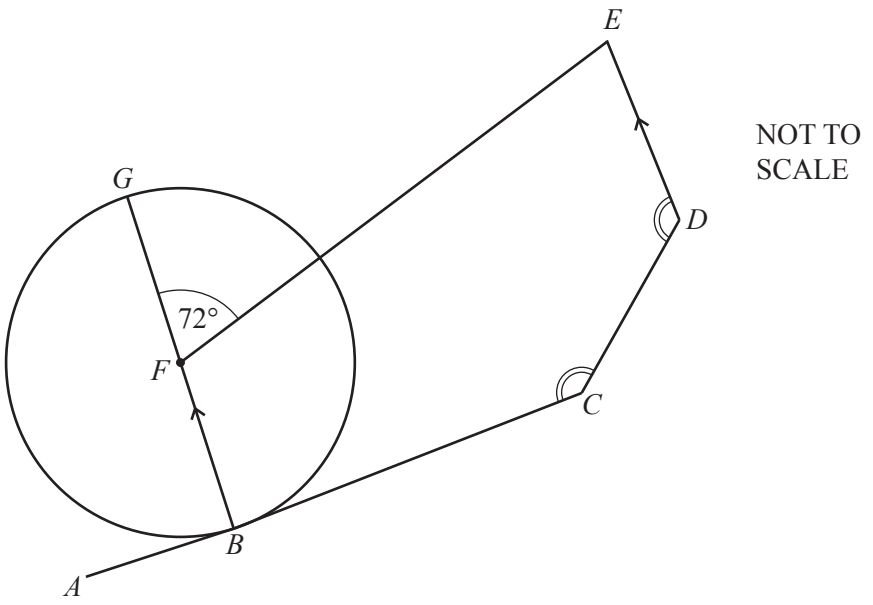
Angle BAC = ... [1]

(ii) angle ABC.

Angle ABC = ... [1]

0580_s18_qp_32_Q8

(b)



The diagram shows a circle, centre F and diameter BG.
AC is a tangent to the circle at B.
BF is parallel to DE, angle $GFE = 72°$ and angle BCD = angle CDE.

(i) Write down the mathematical name of the polygon $BCDEF$.

..... [1]

(ii) Explain why angle FBC is a right angle.

..... [1]

(iii) Find angle BFE, giving a reason for your answer.

Angle BFE =..... because .....

..... [2]

(iv) Find angle FED.

Angle FED = ... [1]

(v) Calculate angle BCD.

Angle BCD = ... [4]